

Assessing the association between oral health and social participation among the elderly people in the UK.

Registration number: [REDACTED]
Word count: 2,718

Abstract

The purpose of this report is to evaluate whether social connection is significant with dental health conditions among the elderly people who are at high risk of dementia in the UK. The dataset is from The English Longitudinal Study of Ageing (ELSA) Wave 7. The independent variables include 2 demographic variables, 2 oral related variables and 3 social related variables. The dependent variable is DentalHealthCondition which is from a self-rated question. There are 3 kinds of tests in this report, one is Chi square Linear-by-Linear Association, another is Spearman Rank Correlation Coefficient, and the other is Logistic regression. The result shows the elderly people's dental health awareness has a significant relationship with other dental health problems and social participation in the UK which means that the quality of oral health can affect the elderly's social participation but it is a mild association.

Introduction

Consuming is an important human need. However, the elderly people with poor oral health such as using dentures do not like to eat out with friends because of public embarrassment (Al-Sultani, 2019). They are tired of taking off their dentures again and again and not being able to masticate. This makes me curious about how oral health affects old people's social participation.

When thinking of the elderly life, some people will also think about dementia. There is no solution for dementia recently, the only thing we can do is to prevent it. The National Health Service official description of how to prevent dementia shows that eating a healthy diet is one of the methods (National Health Service, 2021). In addition, mastication ability has good effects on the brain (Momose et al., 1997; Onozuka, 2002; Hirano, 2008). However, whether they can consume food depends on their dental conditions.

This report will choose some social participation questions and oral health recordings as independent variables to see whether there is association with self-rated dental health conditions.

Literature review

'62% of those aged 65 or over estimated to have dementia have a coded diagnosis of dementia' (the National Health Service, 2021). By the end of 2022, there were 451,992 people diagnosed with dementia, which increased 2.8% from 2021 (the National Health Service, 2022). This trend shows that over half of elderly people potentially have dementia in the UK, and the trend will be increasing. Despite a decline in dementia incidence, the total number of dementia people in the UK is likely to grow because of improved life expectancy (Ahmadi-Abhari, 2017).

The Dementia Action Plan for Wales (LCWG) (2018) recognised dementia as a public health issue. The older people are more likely diagnosed with dementia (Arvanitakis, 2019). Thus, it is important to find factors for dementia. Dementia includes Alzheimer's disease. For reference, I sometimes search for Alzheimer's disease as dementia because it is the most common cause of dementia in the UK. (the National Health Service, 2020).

'Cognitive and social activities were associated with dementia' (Duffner et al., 2022). There is evidence that there is no relationship between mental and physical activities and dementia (Sabia et al., 2017). It is not because dementia causes low mental and physical activities, instead, low activity rate is a sign of dementia before it is diagnosed (Floud et al., 2021). In addition, People who have dementia are admitted to the emergency room higher than people in the same age without dementia (Sommerlad et al., 2019).

Many articles discussed how dementia relates to family and mental health. To be more professional, some articles discussed clinical features for oral. For example, People with dementia have worse oral hygiene such as plaque accumulation and extensive periodontal disease (Daly, 2017). Another finding is that people with mild dementia reduce the time for brushing teeth and dentures which is the reason why they face oral diseases (Gil-Montoya et al., 2017). Therefore, dental care is important for people with dementia (Daly, 2017). However, there is still little literature on the relation between dental health and dementia compared to other features.

On the other hand, it is interesting that dental conditions also affect social participation. For example, people who wear dentures are aware of their friends' perception because they are not able to eat as normal as healthy teeth people (Al-Sultani, 2019). The feature of internalised stigma (Corrigan, 1998) makes people who have SMI (serious mental illness) hide their mental health history in order to avoid rejection from other people they know (Chronister, Chou & Liao, 2013). People who have cognitive problems deliberately avoid socialising in order to protect themselves from social embarrassment which displays 'socially undesirable behaviour' and a feeling of insecurity (Donkers, 2018).

People who live in high HDI (Human Development Index) countries are more likely to discriminate than in low HDI countries (Lasalvia et al., 2015). British people with dementia suffer more stigmatisation than Italian and Polish people (Lion, 2020). This may cause people to not want to go for dementia diagnosis to avoid being treated unfairly.

Research aim and objectives

In this report, the purpose is to understand the elderly people's dental problems, and identify the relation between dental health awareness and social participation among them in the UK.

This report assumes that the participants mostly have a potential risk factor of dementia.

There are 3 objectives:

1. To assess the relation between dental health conditions and removable dentures.
2. To explore if the health conditions depend on demographic variables.
3. To examine whether there is a relation between dental health conditions and social participation.

Methodology

Brief description of the ELSA Wave 7

The English Longitudinal Study of Ageing (ELSA) gives information about the dynamics of health, social, wellbeing and economic circumstances among British people aged 50 and over. The interview started in 2002, and the same group of people will be re-interviewed every 2 years. Some new and younger people engaged in Wave 7 in order to maintain a stable number of people. In Wave 7, the information was collected between June 2014 and May 2015. There were 9,666 people who participated in the survey including 8,249 people who had joined in the first wave. There were some new ideas which broadened the scope of the study. For example, the elderly people were interviewed in more detail about oral health and cognitive function.

Description of 8 used variables

Table 1. The variables

Variable type	Category	Original Variable	Re-coded Variable
Independent	Demographic	Indager	AgeRange
		Disex	Gender
	Oral health	HeDntF & HeDntG	Dentures
		SCACTB	EatOutFrequency
	Social Participation	Scptr7	NotConnectToTheWorld
		scint	InternetUse
		scprtr	LiveWithSomeone
Dependent	Self-check Oral health	HeDntA	DentalHealthCondition

DentalHealthCondition is a self-rated question which can examine the relationship between their real oral situations and their perception of oral health.

Dentures are combined by HeDntF and HeDntG using 'Compute Variable', and the function is 'ANY(?,?)'. Therefore, the value of 0 is including 'No' and 'Missing value'.

EatOutFrequency can be regarded as an oral health type variable, and also social participation type variable.

Old single people are not willing to make new friends in their ages (Goll et al., 2018). This evidence can make LiveWithSomeone as a social participation variable in this report.

Table 2. Overview of variables

Variable	Description	Label	Value
AgeRange	Ages in different range	Aged under 50	1
		Aged from 50 to 59	2
		Aged from 60 to 69	3
		Aged from 70 to 79	4
		Aged from 80 to 89	5
		Aged over 89	6
Gender	Gender	Male	1
		Female	2
Dentures	Do you have dentures or removable false teeth?	No	1
		Yes	2
EatOutFrequency	How often do you eat out of the house?	Never	1
		Less than once a year	2
		About once or twice a year	3
		Every few months	4
		About once a month	5
		Twice a month or more	6
		Missing Value	.
NotConnectToTheWorld	Do you follow up the current event?	I don't have connection with the society	1
		I have connection with the society	2
		Missing Value	.
InternetUse	How often do you use the Internet or email?	Never	1
		Less than every 3 months	2
		At least once every 3 months	3
		At least once a month (but not every week)	4
		At least once a week (but not every day)	5
		Every day, or almost every day	6
		Missing Value	.
LiveWithSomeone	Do you have a husband, wife or partner with whom you live?	No	1
		Yes	2
		Missing Value	.
DentalHealthCondition	The levels of dental health conditions	Poor	1
		Fair	2
		Good	3
		Very good	4
		Excellent	5
		Missing Value	.

Descriptive statistics

All variables are categorical which can be used for qualitative research.

Because there is no suitable regression for ordinal dependent variable. I re-code DentalHealthCondition again into binary so that it can be tested for Logistic regression.

Table 3. Binary dependent variable (re-code from the same variable)

Variable	Description	Label	Value
BiDentalHealthCondition	The levels of dental health conditions (Binary)	Bad	1
		Good	2
		Missing Value	.

Test methods

Chi Square Linear-by-Linear Association

- Learn from week 6 lecture.
- Be used to compare two groups.
- Be used for one binary and one ordinal variables.
- The observations cannot be lower at 20.

Spearman Rank Correlation Coefficient

- Learn from week 7 lecture.
- Be used to explore the association between two ordinal variables.

Logistic regression

- Learn from week 10 lecture.
- The dependent variable should be binary.
- The independent variables can be nominal, ordinal and continuous.
- Be used to assess the relationship between nonparametric variables and a binary dependent variable.

Analytical procedure

There are 4 tests I am going to use.

Test 1

Question: Is there an association between whether you have dentures and self-rated dental health conditions?

The null hypothesis: There is no association between Dentures and DentalHealthCondition.

The alternative hypothesis: There is an association between Dentures and DentalHealthCondition.

Test method: Chi Square Linear-by-Linear Association

Test 2

Question: Is there a correlation between the frequency you eat out of the house and self-rated dental health conditions?

The null hypothesis: There is no correlation between EatOutFrequency and DentalHealthCondition.

The alternative hypothesis: There is a correlation between Dentures and DentalHealthCondition.

Test method: Spearman Rank Correlation Coefficient

Test 3

Question: Is there a relationship between age and self-rated dental health conditions?

The null hypothesis: There is no relationship between AgeRange and DentalHealthCondition.

The alternative hypothesis: There is a relationship between AgeRange and DentalHealthCondition.

Test method: Spearman Rank Correlation Coefficient and splitting the data by Gender

Test 4

Question: Is there a relationship between 'whether you follow up the current event', 'the frequency of using the Internet', 'whether you have a partner living with' and 'self-rated dental health conditions in binary'?

The null hypothesis: There is no relationship between NotConnectToTheWorld, InternetUse, LiveWithSomeone and BiDentalHealthCondition.

The alternative hypothesis: There is a relationship between NotConnectToTheWorld, InternetUse, LiveWithSomeone and BiDentalHealthCondition.

Test method: Logistic regression

Results

Descriptive statistics

Table 4 shows details about the variables. There are 9,666 participants, most of them aged from 60 to 69 (37.6%). Males are a little more than females. 33.6% of valid participants wear dentures while 37.3% of people like to eat out twice a month or more which is also the most percent in EatOutFrequency variable. As to social participation, most of them use the Internet every day, are willing to connect to the world, and have a partner living with them. However, 38.1% of the respondents think their dental health condition is 'good', which presents a neutral attitude.

Table 4. Descriptive statistics

Variable	Type	Label	Value	Frequency	Percent	Valid Percent	Mode
AgeRange	Categorical - Ordinal	Aged under 50	1	175	1.8	1.8	
		Aged from 50 to 59	2	2082	21.5	21.5	
		Aged from 60 to 69	3	3630	37.6	37.6	V
		Aged from 70 to 79	4	2515	26.0	26.0	
		Aged from 80 to 89	5	1071	11.1	11.1	
		Aged over 89	6	193	2.0	2.0	
Gender	Categorical - Binary	Male	1	4298	44.5	44.5	
		Female	2	5368	55.5	55.5	V
Dentures	Categorical - Binary	No	1	6422	66.4	66.4	V
		Yes	2	3244	33.6	33.6	
EatOutFrequency	Categorical - Ordinal	Never	1	387	4.0	4.9	
		Less than once a year	2	195	2.0	2.5	
		About once or twice a year	3	635	6.6	8	
		Every few months	4	1740	18.0	21.9	
		About once a month	5	2019	20.9	25.4	
		Twice a month or more	6	2965	30.7	37.3	V
		Missing Value	.	1725	17.8	-	-
NotConnectToTheWorld	Categorical - Binary	I don't have connection with the society	1	99	1.0	1.2	
		I have connection with the society	2	8009	82.9	98.8	V
		Missing Value	.	1558	16.1	-	-
InternetUse	Categorical - Ordinal	Never	1	1879	19.4	23.5	
		Less than every 3 months	2	112	1.2	1.4	
		At least once every 3 months	3	70	0.7	0.9	
		At least once a month (but not every week)	4	272	2.8	3.4	
		At least once a week (but not every day)	5	899	9.3	11.2	

		Every day, or almost every day	6	4760	49.2	59.6	V
		Missing Value	.	1674	17.3	-	-
LiveWithSomeone	Categorical - Binary	No	1	2161	22.4	26.6	
		Yes	2	5976	61.8	73.4	V
		Missing Value	.	1529	15.8	-	-
DentalHealthCondition	Categorical - Ordinal	Poor	1	666	6.9	6.9	
		Fair	2	1495	15.5	15.5	
		Good	3	3679	38.1	38.1	V
		Very good	4	2686	27.8	27.8	
		Excellent	5	1135	11.7	11.7	
		Missing Value	.	5	0.1	-	-

Inferential Statistics

Test 1

Table 5. Hypothesis

H_0 = There is no significant association between Dentures and DentalHealthCondition.
H_1 = There is a significant association between Dentures and DentalHealthCondition.

			Do you have dentures or removable false teeth?		
			No	Yes	Total
The levels of dental health conditions	Poor	Count	296	370	666
		%	44.40%	55.60%	100.00%
	Fair	Count	818	677	1495
		%	54.70%	45.30%	100.00%
	Good	Count	2367	1312	3679
		%	64.30%	35.70%	100.00%
	Very good	Count	2026	660	2686
		%	75.40%	24.60%	100.00%
	Excellent	Count	911	224	1135
		%	80.30%	19.70%	100.00%
	Total	Count	6418	3243	9661
		%	66.40%	33.60%	100.00%

Table 7. Chi-Square result

	Value	df	Asymptotic Significance (2-sided)
Linear-by-Linear Association	430.436	1	<.001
N of Valid Cases	9661		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 223.56.

According to the result from Table 7, because $p < 0.001$, we can reject the null hypothesis, and accept the alternative hypothesis.

When looking at Table 6, it can be seen that 55.6% of people feel poor of their dental health when they have dentures whereas 80.30% of people feel excellent when they do not have dentures.

Test 2

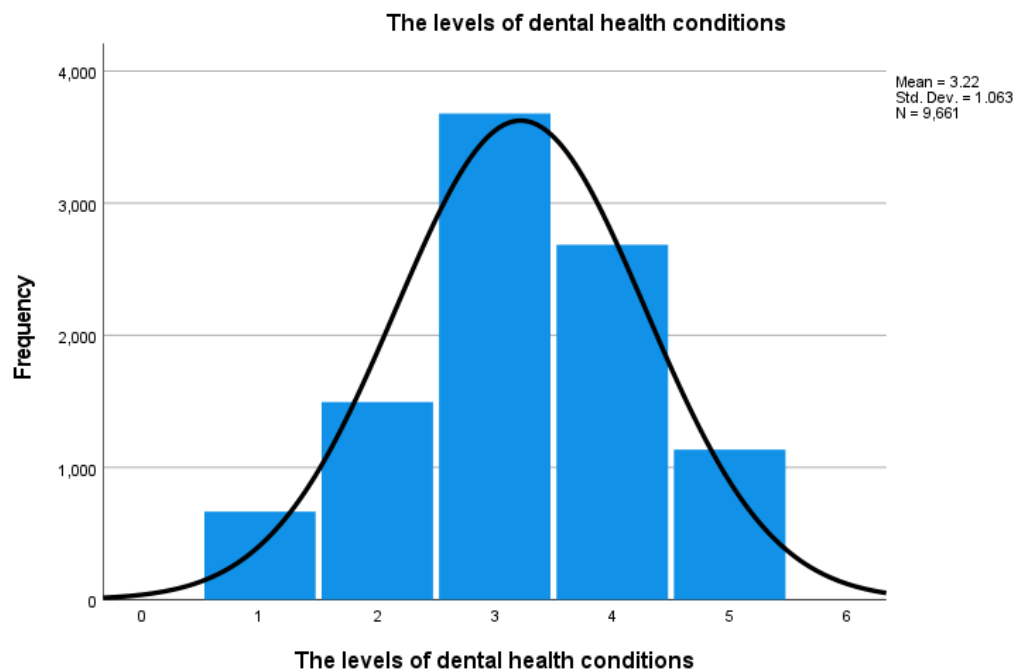
Table 8. Hypothesis

H_0 = There is no correlation between EatOutFrequency and DentalHealthCondition.

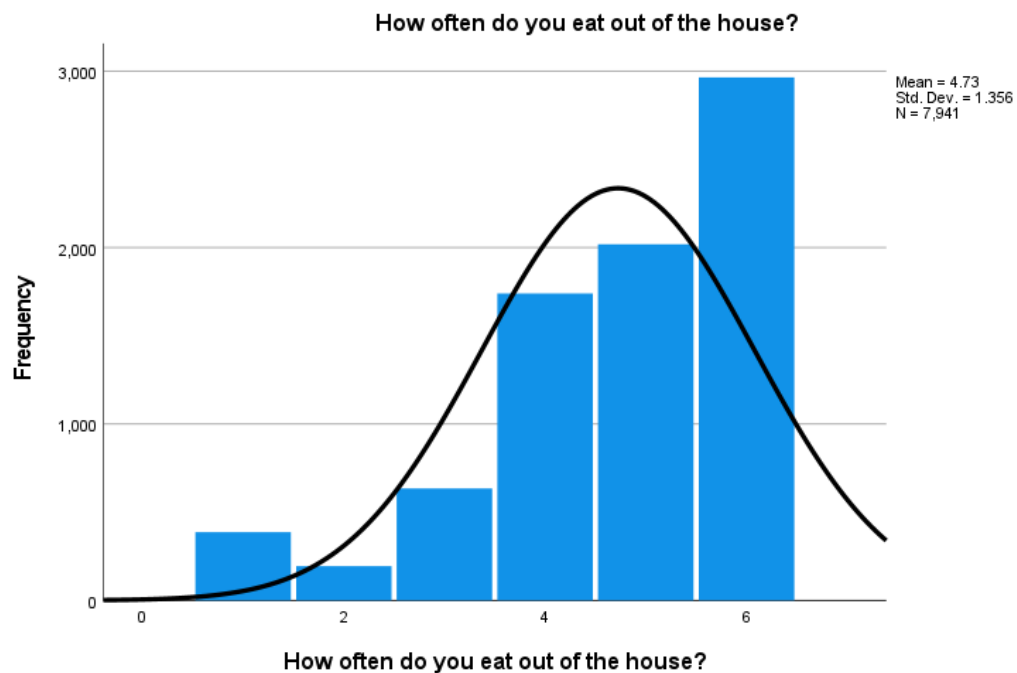
H_1 = There is a correlation between EatOutFrequency and DentalHealthCondition.

Table 9. Statistics		The levels of dental health conditions	How often do you eat out of the house?
N	Valid	9661	7941
	Missing	5	1725
Mean		3.22	4.73
Median		3	5
Mode		3	6
Std. Deviation		1.063	1.356
Skewness		-0.205	-1.098
Std. Error of Skewness		0.025	0.027
Kurtosis		-0.414	0.702
Std. Error of Kurtosis		0.05	0.055
Range		4	5

Graph 1. Histogram of EatOutFrequency



Graph 2. Histogram of DentalHealthCondition



According to graph 1 and 2, it can be seen that both variables are not normally distributed. Graph 1 looks like the tail at the right hand side while graph 2 is significantly skewed distribution.

Table 10. Spearman Rank Correlation Coefficient			The levels of dental health conditions	How often do you eat out of the house?
Spearman's rho	The levels of dental health conditions	Correlation Coefficient	1	.110**
		Sig. (2-tailed)	.	<.001
		N	9661	7939
	How often do you eat out of the house?	Correlation Coefficient	.110**	1
		Sig. (2-tailed)	<.001	.
		N	7939	7941
** Correlation is significant at the 0.01 level (2-tailed).				

The Table 10 displays Sig. (2-tailed) < 0.001. Therefore, we can reject the null hypothesis and accept the alternative hypothesis.

Test 3

Table 11. Hypothesis

H_0 = There is no relationship between AgeRange and DentalHealthCondition.
H_1 = There is a relationship between AgeRange and DentalHealthCondition.

Table 12. Spearman's correlation result				The levels of dental health conditions	Ages in different range
Spearman's rho	Male	The levels of dental health conditions	Correlation Coefficient	1	-.039*
			Sig. (2-tailed)	.	0.011
			N	4296	4296
		Ages in different range	Correlation Coefficient	-.039*	1
			Sig. (2-tailed)	0.011	.
			N	4296	4298
	Female	The levels of dental health conditions	Correlation Coefficient	1	-.084**
			Sig. (2-tailed)	.	<.001
			N	5365	5365
		Ages in different range	Correlation Coefficient	-.084**	1
			Sig. (2-tailed)	<.001	.
			N	5365	5368
* Correlation is significant at the 0.05 level (2-tailed).					
** Correlation is significant at the 0.01 level (2-tailed).					

When looking at Table 12, the data is splitted by Gender. The p value on Male and Female are both lower at 0.05, which are 0.011 and smaller than 0.001, respectively. As a result, we can reject the null hypothesis and accept the alternative hypothesis.

Test 4

Table 13. Hypothesis

H_0 = There is no relationship between NotConnectToTheWorld, InternetUse, LiveWithSomeone and BiDentalHealthCondition.
H_1 = There is a relationship between NotConnectToTheWorld, InternetUse, LiveWithSomeone and BiDentalHealthCondition.

Table 14. Case Processing Summary

Unweighted Cases ^a		N	Percent
Selected Cases	Included in Analysis	7925	82
	Missing Cases	1741	18
	Total	9666	100
Unselected Cases		0	0
Total		9666	100
a. If weight is in effect, see classification table for the total number of cases.			

Referring to Table 14, there are 1,741 missing values which means almost one fifth participants were not involved in this test.

Table 15. Categorical Variables Codings

		Frequency	Parameter coding				
			(1)	(2)	(3)	(4)	(5)
How often do you use the Internet or email?	Never	1848	0.000	0.000	0.000	0.000	0.000
	Less than every 3 months	112	1.000	0.000	0.000	0.000	0.000
	At least once every 3 months	69	0.000	1.000	0.000	0.000	0.000
	At least once a month (but not every week)	271	0.000	0.000	1.000	0.000	0.000
	At least once a week (but not every day)	892	0.000	0.000	0.000	1.000	0.000
	Every day, or almost every day	4733	0.000	0.000	0.000	0.000	1.000
Do you follow up the current event?	I don't have connection with the society	94	0.000				
	I have connection with the society	7831	1.000				
Do you have a husband, wife or partner with whom	No	2087	0.000				

you live?	Yes	5838	1.000				
-----------	-----	------	-------	--	--	--	--

I identify 'Never' as the reference label in the InternetUse variable which showed on Table 15. I also put 'I don't have connection with the society' and 'No' as the reference labels for other independent variables.

Table 16. Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 0	Constant	1.311	0.027	2278.986	1	.000	3.712

Table 17. Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	8139.572 ^a	0.007	0.010

a. Estimation terminated at iteration number 4 because parameter estimates changed by less than .001.

Table 18. Variables in the Equation

								95% C.I. for EXP(B)	
		B	S.E.	Wald	df	Sig.	Exp(B)	Lower	Upper
Step 1	How often do you use the Internet or email?			24.659	5	<.001			
	How often do you use the Internet or email?(1)	-0.288	0.213	1.831	1	0.176	0.750	0.494	1.138
	How often do you use the Internet or email?(2)	0.187	0.297	0.396	1	0.529	1.206	0.673	2.159
	How often do you use the Internet or email?(3)	0.247	0.161	2.362	1	0.124	1.280	0.934	1.755
	How often do you use the Internet or email?(4)	0.150	0.098	2.325	1	0.127	1.161	0.958	1.408
	How often do you use the Internet or email?(5)	0.292	0.067	18.812	1	<.001	1.339	1.174	1.528
	Do you have a husband, wife or partner with whom you live?(1)	0.224	0.062	12.981	1	<.001	1.251	1.108	1.414
	Do you follow up the current event?(1)	0.402	0.224	3.234	1	0.072	1.495	0.964	2.319
	Constant	0.562	0.219	6.571	1	0.010	1.754		

a. Variable(s) entered on step 1: How often do you use the Internet or email?, Do you have a husband, wife or partner with whom you live?, Do you follow up the current event?.

According to Table 16, the p value is smaller than 0.001 so that we can reject the null hypothesis and accept the alternative hypothesis. In addition, the result of Nagelkerke R Square on Table 17 shows that around 1% of the variability in people's feeling of dental health can be explained by these independent variables.

Although the overall frequency of people using the Internet or email is significant ($p < 0.001$) for people's dental conditions, only the value of using the Internet or email every day is significant and with positive relationships ($\text{Exp}(B) = 1.339$). The independent variable of living with a partner is also significant to the dental health conditions whereas the variable of people following the current events is not significant, with the p value being around 0.072.

Discussion

In this report, I did not put any dementia diagnosis variables such as `hedbdde` and `hedbsde` as dependent variables. It is because the missing values in the variables account for a big part. For example, 'Whether confirms dementia diagnosis' has 9625 (99.6%) missing values, and there are only 40 people confirmed that they had dementia. This may lead to the result incorrectly. Therefore, according to NHS data (2022), there are potentially over a half of people who are aged 65 or over diagnosed as dementia, I assume that most people who participated in ELSA Wave 7 had a risk factor of dementia.

The relation between dental health conditions and removable dentures is significant (Table 7), and we can explain that people who wear dentures feel less satisfied with their dental condition. This result can be related to Gil-Montoya et al. (2017) mentioned that people with mild dementia do not have much time washing their teeth and dentures and this results in oral diseases.

As to the second test for testing the relationship between frequency of eating out and dental health conditions, they have a smaller relationship. Although the correlation coefficient is 0.110 which is positive association, it is modestly correlated (Table 10). The reason why they are not strongly associated together could be because around 37.3% (valid percent) of people liked to eat out twice a month or more which may be opposed for a big proportion of people who wear dentures.

When exploring the test 3 on Table 12, p values of both genders are smaller than 0.05 which can reject the null hypothesis. The correlation coefficients on both male and female are negative association, with -0.039 and -0.084 respectively, which means that the older people think that they feel their dental health conditions is worse than the middle aged people. There is modest correlation between age and dental health conditions, but it can be seen that females(-0.084) have more association than males(-0.039).

When looking at the test 4 result (Table 16), although the overall logistic regression shows dental health conditions (binary) is significant association with social participation (`NotConnectToTheWorld`, `InternetUse` and `LiveWithSomeone`), if we look more detailed on Table 18, there are not all independent variables showing significant. Only the independent variable of who lives with someone (`LiveWithSomeone`) is significant to the dental health conditions. This result can be explained that social participation is significant but has little association with dental health conditions.

The biggest limitation of this report is too little useful information about dementia. Although there are some questions about dementia and Alzheimer's disease, the responses are not able to be qualified because of a big part of missing values. Moreover, many questions for social connection are multiple options, and the editors put the multiple options as variables separately. This means that if you choose one of these variables, you cannot receive many respondents, and you have to understand the variables carefully and re-code correctly unless you have the wrong result and interpretation.

Because there is a limited time to do this report. There are some suggestions for further research. One is to add more dental variables in the test such as the number of natural teeth and whether mouth or teeth feel painful in order to assess whether their self-rated dental health conditions are associated with their actual dental problems. On the other hand, social related measurements are too wide to cover all. Defining more narrow social variables as independent variables can make the evaluation more persuasive. For example, the topic can be narrated to activity participation, internet society or family support.

Conclusion

How people consume depends on their quality of oral hygiene. However, there are not many articles proving the relationship between oral hygiene and social connection. Some journal articles I reviewed only talked about the feeling of unhealthy teeth (Al-Sultani, 2019). This report aimed to explore the relationship between oral health conditions and social participation. The results from a series of tests showed that they are significantly associated. For example, people who wear dentures feel more badly about their dental conditions. Specifically, the females have more association than males. However, although it is significant for the relationship between self-rated dental health conditions and social participation, their logistic regression showed that their correlation is relatively low. The further research for this report can add more dental variables to emphasise the relationship between real oral conditions and self-feeling, or narrow the topic to focus on one situation of social issues.

Reference

- Al-Sultani, H. F., Field, J. C., Thomason, J., & Moynihan, P. J. (2019). The impact of replacement conventional dentures on eating experience. *JDR Clinical & Translational Research*, 4(1), 29-40.
- National Health Service. (2021, June 26). *Can dementia be prevented?*. <https://www.nhs.uk/conditions/dementia/dementia-prevention/>
- Momose, T., Nishikawa, J., Watanabe, T., Sasaki, Y., Senda, M., Kubota, K., ... & Minakuchi, S. (1997). Effect of mastication on regional cerebral blood flow in humans examined by positron-emission tomography with 15O-labelled water and magnetic resonance imaging. *Archives of oral biology*, 42(1), 57-61.
- Onozuka, M., Fujita, M., Watanabe, K., Hirano, Y., Niwa, M., Nishiyama, K., & Saito, S. (2002). Mapping brain region activity during chewing: a functional magnetic resonance imaging study. *Journal of Dental Research*, 81(11), 743-746.
- Hirano, Y., Obata, T., Kashikura, K., Nonaka, H., Tachibana, A., Ikehira, H., & Onozuka, M. (2008). Effects of chewing in working memory processing. *Neuroscience Letters*, 436(2), 189-192.
- Arvanitakis, Z., & Bennett, D. A. (2019). What is dementia?. *Jama*, 322(17), 1728-1728.
- National Health Service. (2021, September 16). *Recorded Dementia Diagnoses August 2021*. <https://digital.nhs.uk/data-and-information/publications/statistical/recorded-dementia-diagnoses/august-2021>
- National Health Service. (2022, December 26). *NHS launches new dementia diagnosis drive*. <https://www.england.nhs.uk/2022/12/nhs-launches-new-dementia-diagnosis-drive/>
- Ahmadi-Abhari, S., Guzman-Castillo, M., Bandosz, P., Shipley, M. J., Muniz-Terrera, G., Singh-Manoux, A., ... & Brunner, E. J. (2017). Temporal trend in dementia incidence since 2002 and projections for prevalence in England and Wales to 2040: modelling study. *bmj*, 358.
- LCWG, Llywodraeth Cymru Welsh Government. (2018). *Dementia action plan for Wales*. GOV.Wales. <https://www.gov.wales/sites/default/files/publications/2019-04/dementia-action-plan-for-wales.pdf>
- National Health Service. (2021, July 5). *Alzheimer's disease*. <https://www.nhs.uk/conditions/alzheimers-disease/>
- Floud, S., Balkwill, A., Sweetland, S., Brown, A., Reus, E. M., Hofman, A., ... & Beral, V. (2021). Cognitive and social activities and long-term dementia risk: the prospective UK Million Women Study. *The Lancet Public Health*, 6(2), e116-e123.
- Duffner, L. A., Deckers, K., Cadar, D., Steptoe, A., De Vugt, M., & Köhler, S. (2022). The role of cognitive and social leisure activities in dementia risk: assessing longitudinal associations of modifiable and on-modifiable risk factors. *Epidemiology and psychiatric sciences*, 31.
- Sommerlad, A., Perera, G., Mueller, C., Singh-Manoux, A., Lewis, G., Stewart, R., & Livingston, G. (2019). Hospitalisation of people with dementia: evidence from English electronic health records from 2008 to 2016. *European journal of epidemiology*, 34(6), 567-577.
- Daly, B., Thompsell, A., Sharpling, J., Rooney, Y. M., Hillman, L., Wanyonyi, K. L., ... & Gallagher, J. E. (2017). Evidence summary: the relationship between oral health and dementia. *British dental journal*, 223(11), 846-853.
- Gil-Montoya, J. A., Sánchez-Lara, I., Carnero-Pardo, C., Fornieles-Rubio, F., Montes, J., Barrios, R., ... & Bravo, M. (2017). Oral hygiene in the elderly with different degrees of cognitive impairment and dementia. *Journal of the American Geriatrics Society*, 65(3), 642-647.

- Corrigan, P. W. (1998). The impact of stigma on severe mental illness. *Cognitive and behavioral practice*, 5(2), 201-222.
- Chronister, J., Chou, C. C., & Liao, H. Y. (2013). The role of stigma coping and social support in mediating the effect of societal stigma on internalised stigma, mental health recovery, and quality of life among people with serious mental illness. *Journal of Community Psychology*, 41(5), 582-600.
- Donkers, H. W. (2018). *Social participation dilemmas in dementia* (Doctoral dissertation, [SI: sn]).
- Lasalvia, A., Van Bortel, T., Bonetto, C., Jayaram, G., Van Weeghel, J., Zoppei, S., ... & Aspen/Indigo Study Group. (2015). Cross-national variations in reported discrimination among people treated for major depression worldwide: the ASPEN/INDIGO international study. *The British Journal of Psychiatry*, 207(6), 507-514.
- Lion, K. M., Szcześniak, D., Bulińska, K., Evans, S. B., Evans, S. C., Saibene, F. L., ... & Rymaszewska, J. (2020). Do people with dementia and mild cognitive impairments experience stigma? A cross-cultural investigation between Italy, Poland and the UK. *Aging & Mental Health*, 24(6), 947-955.
- Goll, J. C., Charlesworth, G., Scior, K., & Stott, J. (2018). Correction: Barriers to social participation among lonely older adults: The influence of social fears and identity. *Plos one*, 13(7), e0201510.

Appendix

Frequencies

		Statistics							
		Gender	Ages in different range	Do you have a denture or removable false teeth?	How often do you eat out of the house?	Do you follow up the current event?	How often do you use the Internet or email?	Do you have a husband, wife or partner with whom you live?	The levels of dental health conditions
N	Valid	9666	9666	9666	7941	8108	7992	8137	9661
	Missing	0	0	0	1725	1558	1674	1529	5
Mode		2	3	1	6	2	6	2	3

Frequency Table

Gender					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	4298	44.5	44.5	44.5
	Female	5368	55.5	55.5	100.0
	Total	9666	100.0	100.0	

Ages in different range					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Aged under 50	175	1.8	1.8	1.8
	Aged from 50 to 59	2082	21.5	21.5	23.3
	Aged from 60 to 69	3630	37.6	37.6	60.9
	Aged from 70 to 79	2515	26.0	26.0	86.9
	Aged from 80 to 89	1071	11.1	11.1	98.0
	Aged over 89	193	2.0	2.0	100.0
	Total	9666	100.0	100.0	

Do you have a denture or removable false teeth?					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	6422	66.4	66.4	66.4
	Yes	3244	33.6	33.6	100.0
	Total	9666	100.0	100.0	

How often do you eat out of the house?					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Never	387	4.0	4.9	4.9
	Less than once a year	195	2.0	2.5	7.3
	About once or twice a year	635	6.6	8.0	15.3
	Every few months	1740	18.0	21.9	37.2
	About once a month	2019	20.9	25.4	62.7
	Twice a month or more	2965	30.7	37.3	100.0

	Total	7941	82.2	100.0	
Missing	0	1725	17.8		
Total		9666	100.0		

Do you follow up the current event?

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	I dont have connection with the society	99	1.0	1.2	1.2
	I have connection with the society	8009	82.9	98.8	100.0
	Total	8108	83.9	100.0	
Missing	System	1558	16.1		
Total		9666	100.0		

How often do you use the Internet or email?

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Never	1879	19.4	23.5	23.5
	Less than every 3 months	112	1.2	1.4	24.9
	At least once every 3 months	70	.7	.9	25.8
	At least once a month (but not every week)	272	2.8	3.4	29.2
	At least once a week (but not every day)	899	9.3	11.2	40.4
	Every day, or almost every day	4760	49.2	59.6	100.0
	Total	7992	82.7	100.0	
Missing	0	1674	17.3		
Total		9666	100.0		

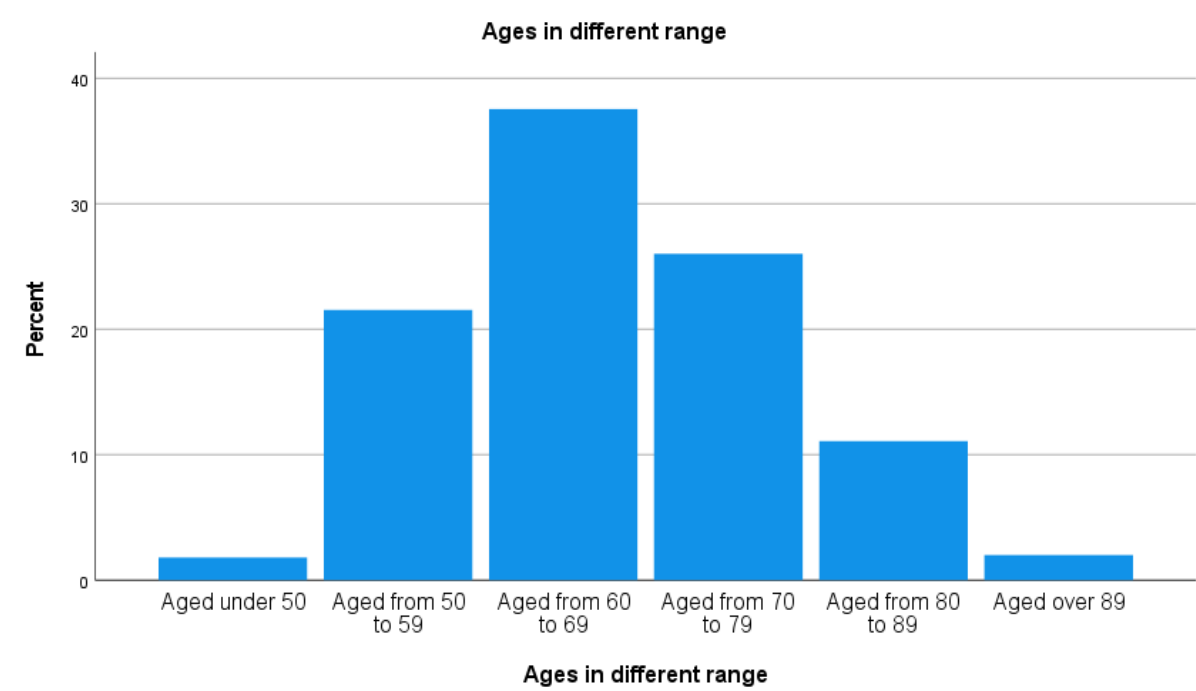
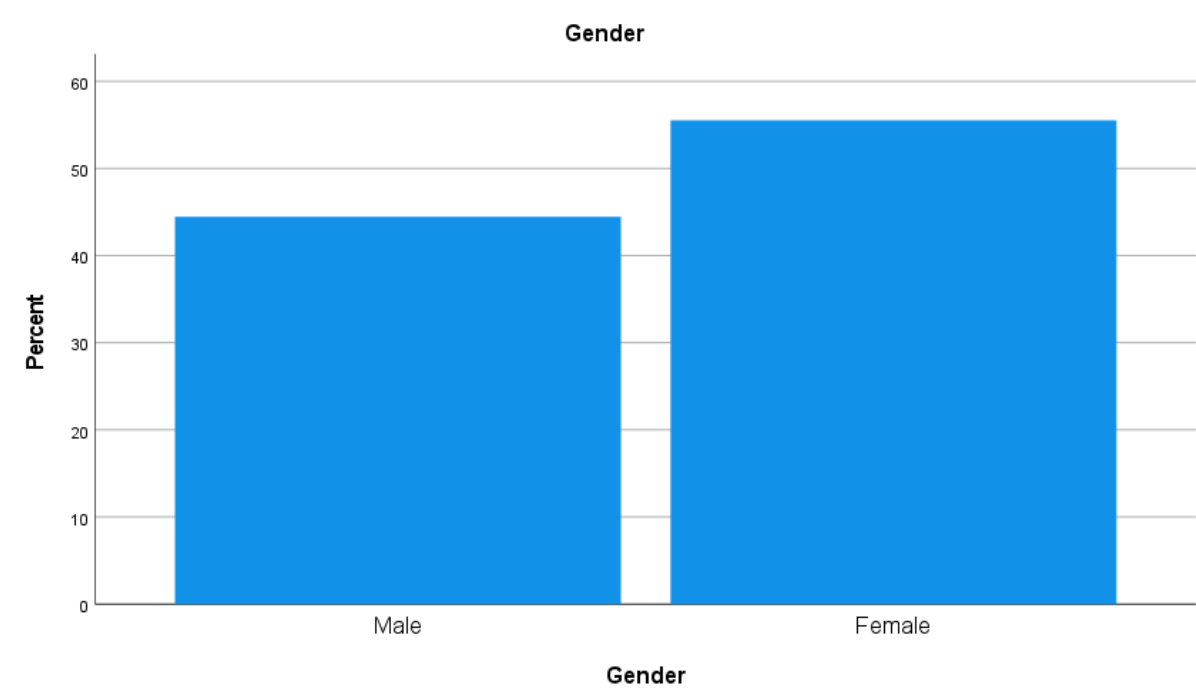
Do you have a husband, wife or partner with whom you live?

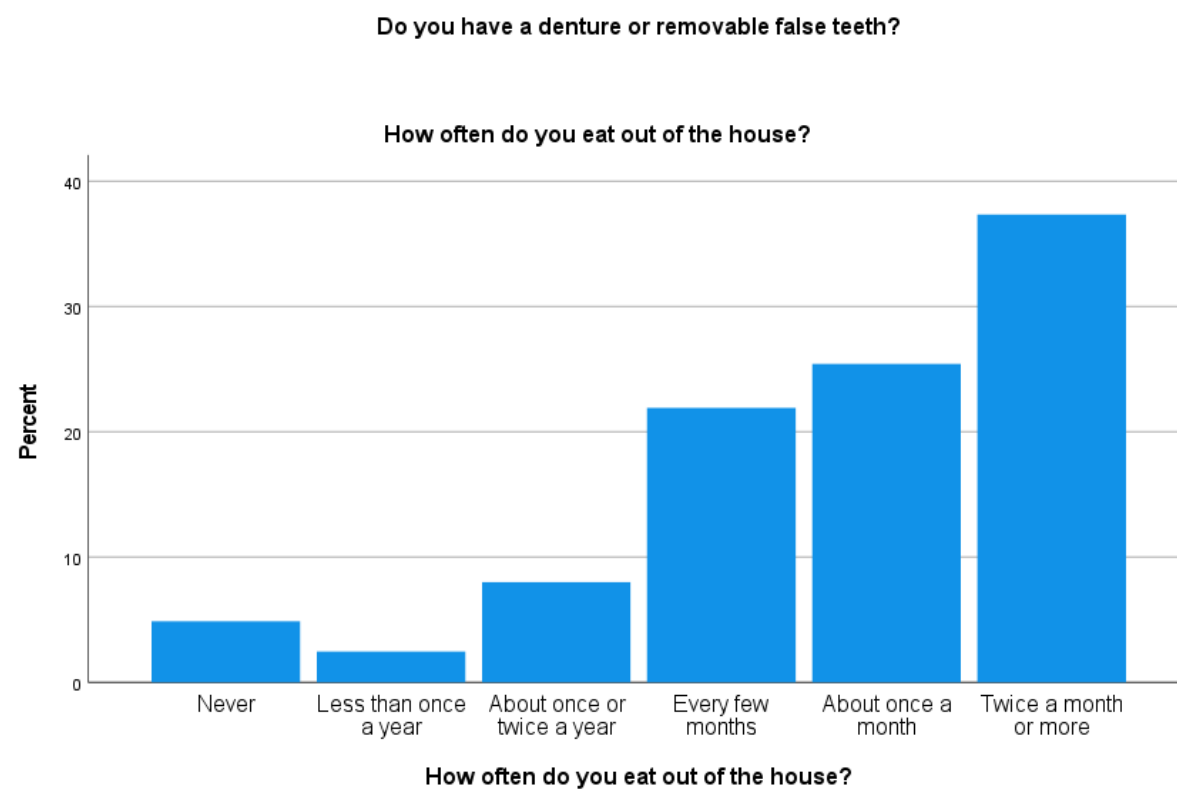
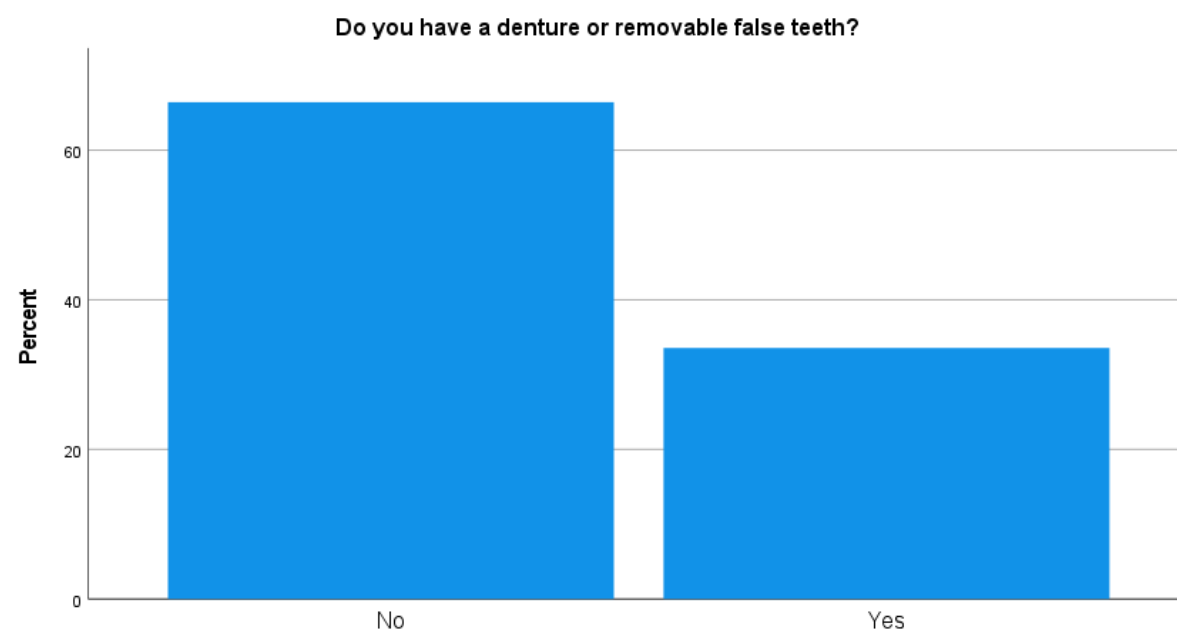
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	2161	22.4	26.6	26.6
	Yes	5976	61.8	73.4	100.0
	Total	8137	84.2	100.0	
Missing	0	1529	15.8		
Total		9666	100.0		

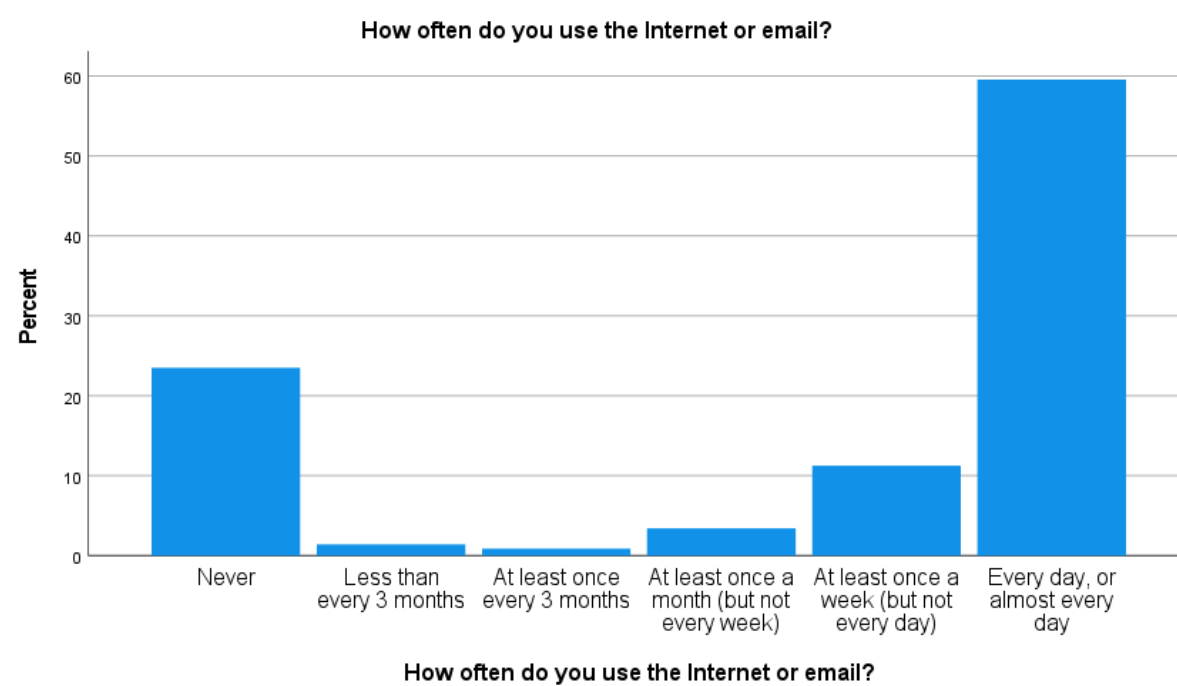
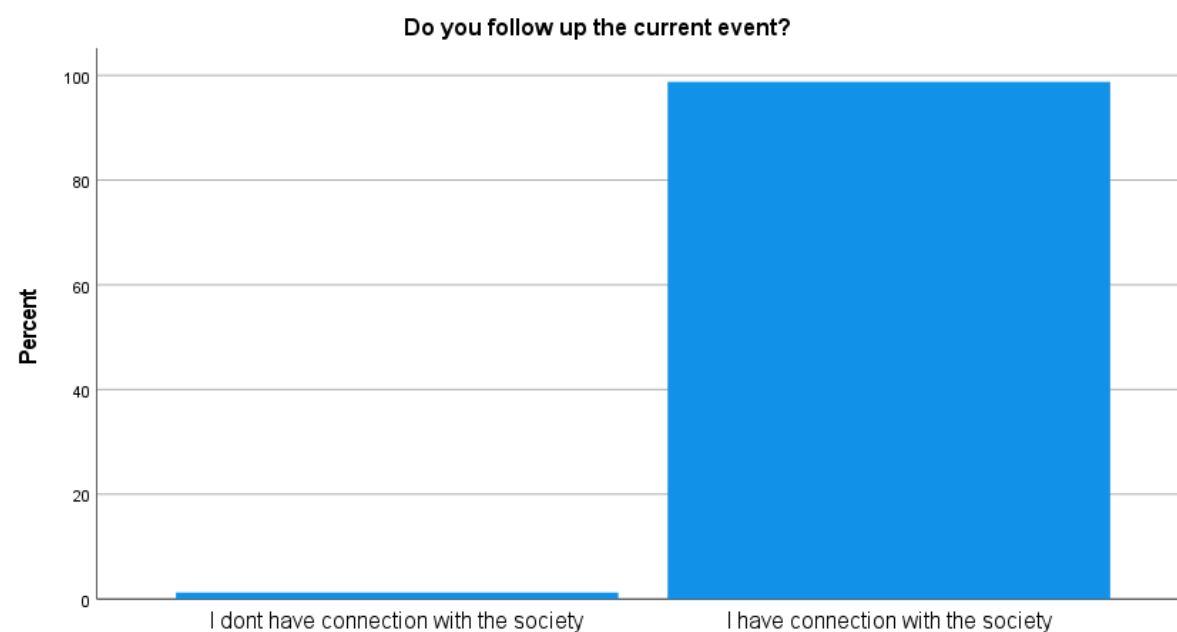
The levels of dental health conditions

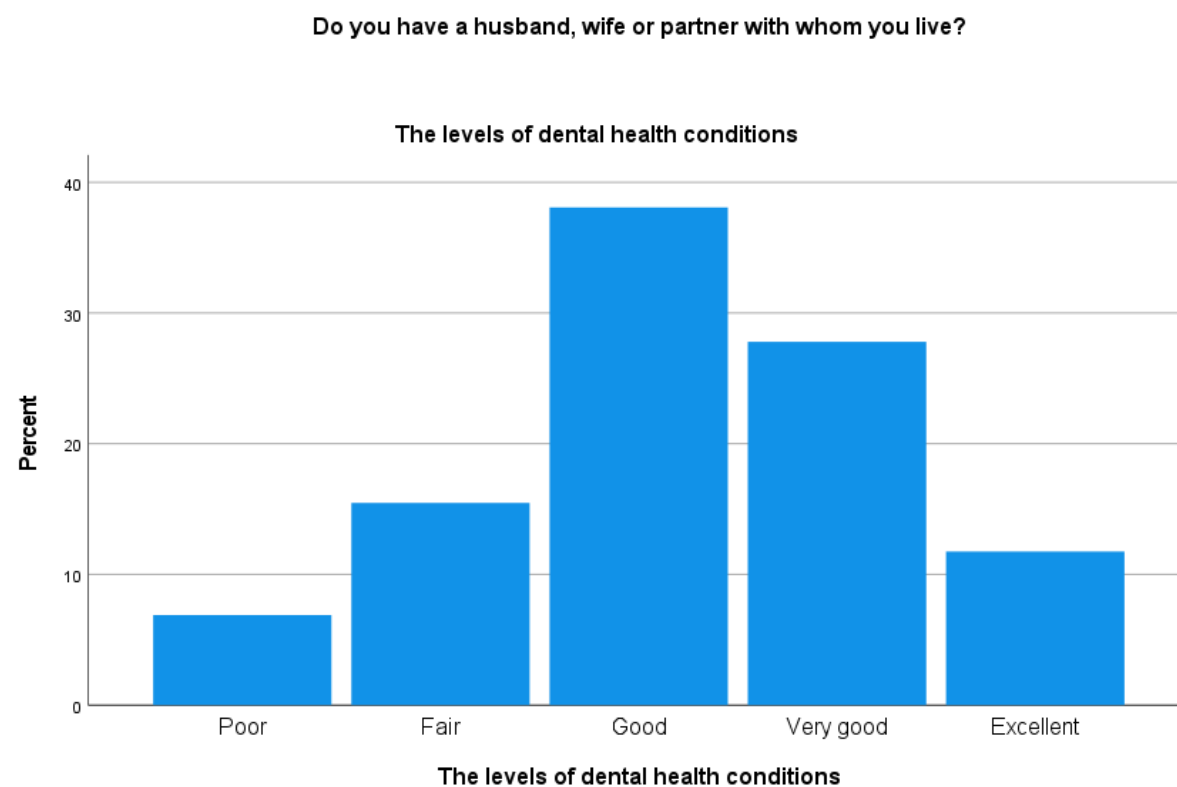
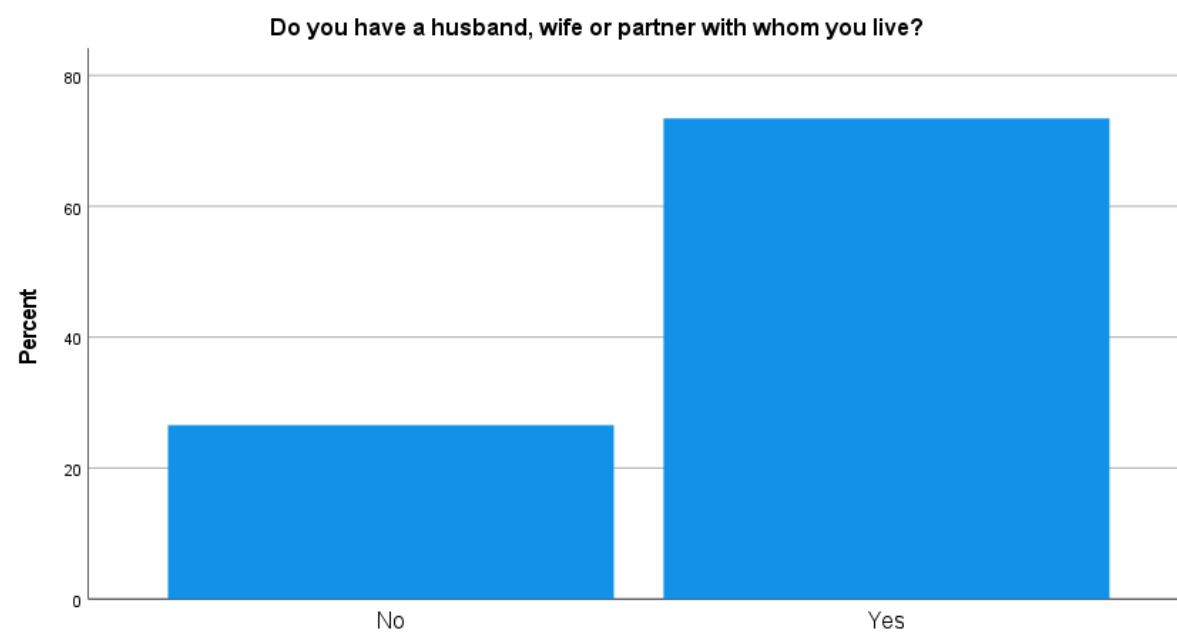
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Poor	666	6.9	6.9	6.9
	Fair	1495	15.5	15.5	22.4
	Good	3679	38.1	38.1	60.4
	Very good	2686	27.8	27.8	88.3
	Excellent	1135	11.7	11.7	100.0
	Total	9661	99.9	100.0	
Missing	System	5	.1		
Total		9666	100.0		

Bar Chart









Crosstabs

Case Processing Summary

	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
The levels of dental health conditions * Do you have a denture or removable false teeth?	9661	99.9%	5	0.1%	9666	100.0%

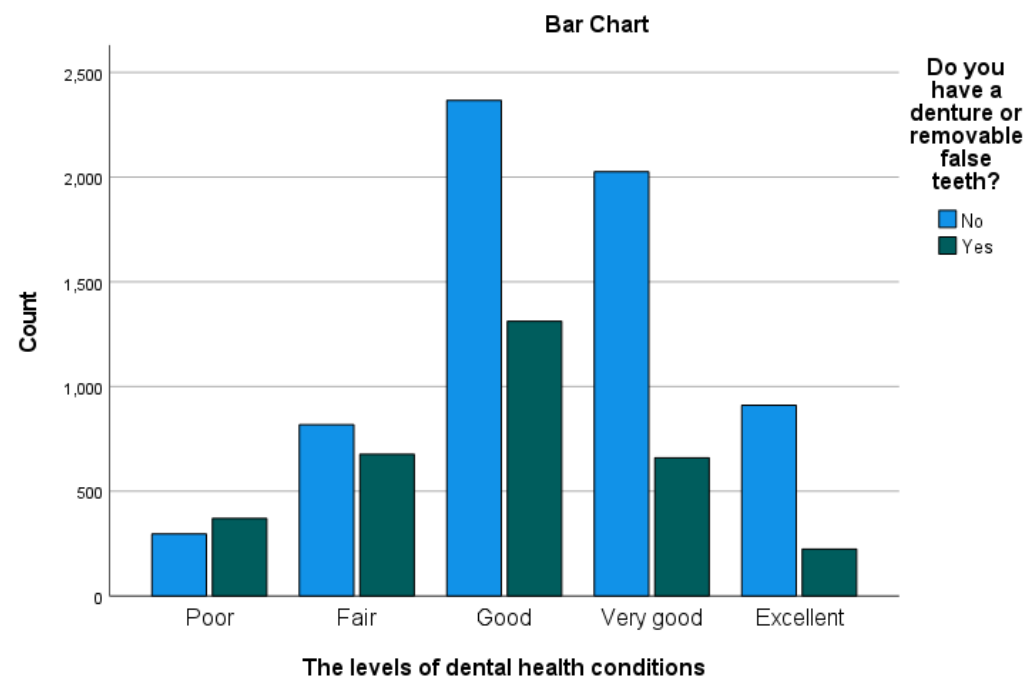
The levels of dental health conditions * Do you have a denture or removable false teeth?
Crosstabulation

			Do you have a denture or removable false teeth?		Total
			No	Yes	
The levels of dental health conditions	Poor	Count	296	370	666
		% within The levels of dental health conditions	44.4%	55.6%	100.0%
	Fair	Count	818	677	1495
		% within The levels of dental health conditions	54.7%	45.3%	100.0%
	Good	Count	2367	1312	3679
		% within The levels of dental health conditions	64.3%	35.7%	100.0%
	Very good	Count	2026	660	2686
		% within The levels of dental health conditions	75.4%	24.6%	100.0%
	Excellent	Count	911	224	1135
		% within The levels of dental health conditions	80.3%	19.7%	100.0%
Total		Count	6418	3243	9661
		% within The levels of dental health conditions	66.4%	33.6%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	438.510 ^a	4	<.001
Likelihood Ratio	439.454	4	<.001
Linear-by-Linear Association	430.436	1	<.001
N of Valid Cases	9661		

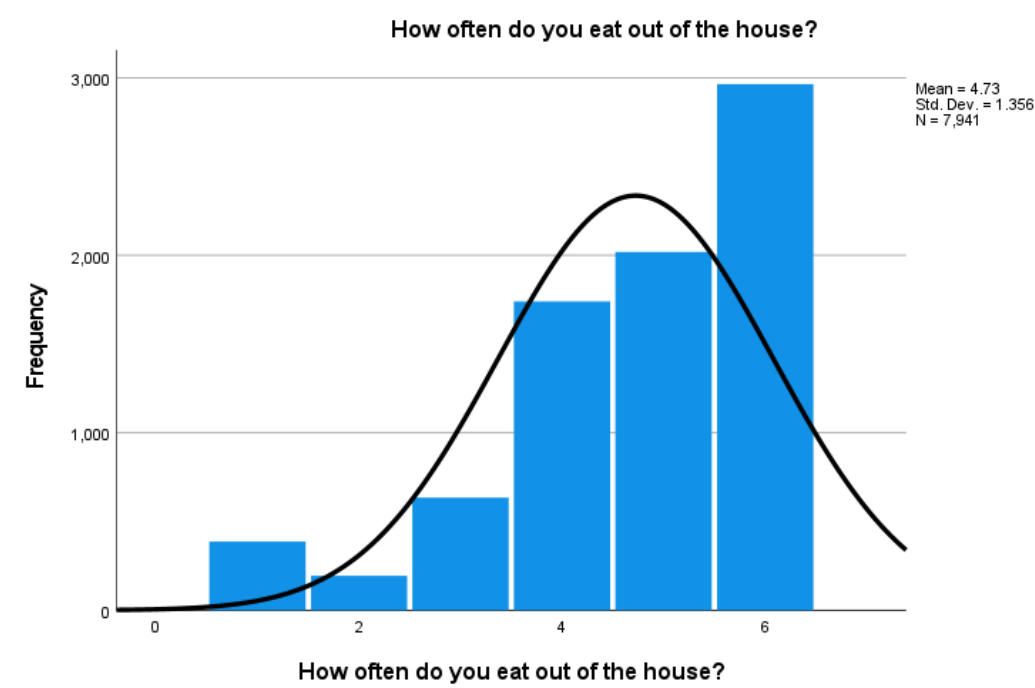
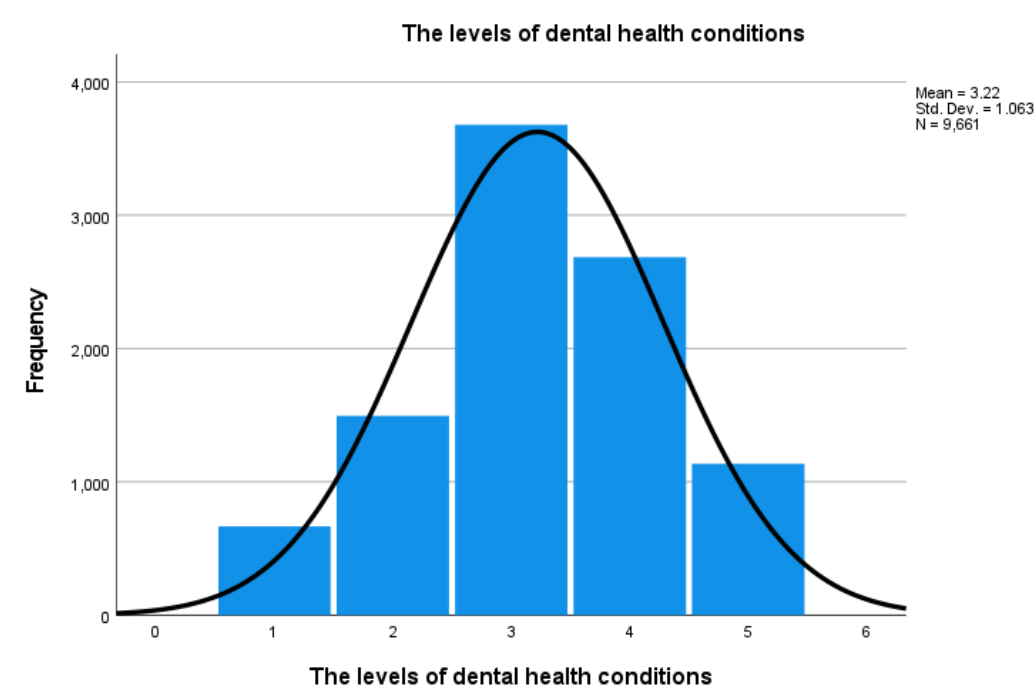
a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 223.56.



Frequencies

		Statistics	
		The levels of dental health conditions	How often do you eat out of the house?
N	Valid	9661	7941
	Missing	5	1725
Mean		3.22	4.73
Median		3.00	5.00
Mode		3	6
Std. Deviation		1.063	1.356
Skewness		-.205	-1.098
Std. Error of Skewness		.025	.027
Kurtosis		-.414	.702
Std. Error of Kurtosis		.050	.055
Range		4	5

Histogram



Nonparametric Correlations

Correlations			The levels of dental health conditions	How often do you eat out of the house?
Spearman's rho	The levels of dental health conditions	Correlation Coefficient	1.000	.110**
		Sig. (2-tailed)	.	<.001
		N	9661	7939
	How often do you eat out of the house?	Correlation Coefficient	.110**	1.000
		Sig. (2-tailed)	<.001	.
		N	7939	7941

** . Correlation is significant at the 0.01 level (2-tailed).

Nonparametric Correlations

Correlations				The levels of dental health conditions	Ages in different range
Spearman's rho	Gender				
	Male	The levels of dental health conditions	Correlation Coefficient	1.000	-.039*
			Sig. (2-tailed)	.	.011
			N	4296	4296
		Ages in different range	Correlation Coefficient	-.039*	1.000
			Sig. (2-tailed)	.011	.
			N	4296	4298
	Female	The levels of dental health conditions	Correlation Coefficient	1.000	-.084**
			Sig. (2-tailed)	.	<.001
			N	5365	5365
		Ages in different range	Correlation Coefficient	-.084**	1.000
			Sig. (2-tailed)	<.001	.
			N	5365	5368

*. Correlation is significant at the 0.05 level (2-tailed).

** . Correlation is significant at the 0.01 level (2-tailed).

Logistic Regression

Case Processing Summary

Unweighted Cases ^a		N	Percent
Selected Cases	Included in Analysis	7925	82.0
	Missing Cases	1741	18.0
	Total	9666	100.0
Unselected Cases		0	.0
Total		9666	100.0

a. If weight is in effect, see classification table for the total number of cases.

Dependent Variable
Encoding

Original Value	Internal Value
Bad	0
Good	1

Categorical Variables Codings

		Frequency	(1)	(2)	Parameter coding		
					(3)	(4)	(5)
How often do you use the Internet or email?	Never	1848	.000	.000	.000	.000	.000
	Less than every 3 months	112	1.000	.000	.000	.000	.000
	At least once every 3 months	69	.000	1.000	.000	.000	.000
	At least once a month (but not every week)	271	.000	.000	1.000	.000	.000
	At least once a week (but not every day)	892	.000	.000	.000	1.000	.000
	Every day, or almost every day	4733	.000	.000	.000	.000	1.000
Do you follow up the current event?	I dont have connection with the society	94	.000				
	I have connection with the society	7831	1.000				
Do you have a husband, wife or partner with whom you live?	No	2087	.000				
	Yes	5838	1.000				

Block 0: Beginning Block

Classification Table^{a,b}

			Predicted The levels of dental health conditions (Binary)		Percentage Correct
Observed			Bad	Good	
Step 0	The levels of dental health conditions (Binary)	Bad	0	1682	.0
		Good	0	6243	100.0
	Overall Percentage				78.8

a. Constant is included in the model.

b. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 0	Constant	1.311	.027	2278.986	1	.000	3.712

Variables not in the Equation

			Score	df	Sig.
Step 0	Variables	How often do you use the Internet or email?	37.610	5	<.001
		How often do you use the Internet or email?(1)	5.668	1	.017
		How often do you use the Internet or email?(2)	.011	1	.916
		How often do you use the Internet or email?(3)	.145	1	.704
		How often do you use the Internet or email?(4)	.244	1	.621
		How often do you use the Internet or email?(5)	27.447	1	<.001
		Do you have a husband, wife or partner with whom you live?(1)	24.314	1	<.001
		Do you follow up the current event?(1)	9.349	1	.002
	Overall Statistics		55.731	7	<.001

Block 1: Method = Enter

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	53.438	7	<.001
	Block	53.438	7	<.001
	Model	53.438	7	<.001

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	8139.572 ^a	.007	.010

a. Estimation terminated at iteration number 4 because parameter estimates changed by less than .001.

Classification Table^a

			Predicted		Percentage Correct
			The levels of dental health conditions (Binary)		
Observed			Bad	Good	
Step 1	The levels of dental health conditions (Binary)	Bad	0	1682	.0
		Good	0	6243	100.0
	Overall Percentage				78.8

a. The cut value is .500

Variables in the Equation									
		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I. for EXP(B)	
								Lower	Upper
Step 1 ^a	How often do you use the Internet or email?			24.659	5	<.001			
	How often do you use the Internet or email?(1)	-.288	.213	1.831	1	.176	.750	.494	1.138
	How often do you use the Internet or email?(2)	.187	.297	.396	1	.529	1.206	.673	2.159
	How often do you use the Internet or email?(3)	.247	.161	2.362	1	.124	1.280	.934	1.755
	How often do you use the Internet or email?(4)	.150	.098	2.325	1	.127	1.161	.958	1.408
	How often do you use the Internet or email?(5)	.292	.067	18.812	1	<.001	1.339	1.174	1.528
	Do you have a husband, wife or partner with whom you live?(1)	.224	.062	12.981	1	<.001	1.251	1.108	1.414
	Do you follow up the current event?(1)	.402	.224	3.234	1	.072	1.495	.964	2.319
	Constant	.562	.219	6.571	1	.010	1.754		

a. Variable(s) entered on step 1: How often do you use the Internet or email?, Do you have a husband, wife or partner with whom you live?, Do you follow up the current event?.

Frequencies

Statistics

Whether confirms dementia diagnosis

N	Valid	9666
	Missing	0

Whether confirms dementia diagnosis

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Item not applicable	9625	99.6	99.6	99.6
	Yes	40	.4	.4	100.0
	No	1	.0	.0	100.0
	Total	9666	100.0	100.0	